TIC 224288919

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        2.928493 d
  Epoch (t0)    3341.4679 BTJD
  Duration      0.1700 d
  Depth         0.057 %  (570 ppm)
  SNR           15.3

INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      4  (gate >= 3)
  Robust depth ratio     0.98  (gate >= 0.50)
  Robust depth sigma     10.5  (gate >= 7.0)
  Odd/even z-score       0.55  (consistent)
  Secondary eclipse      not detected
```

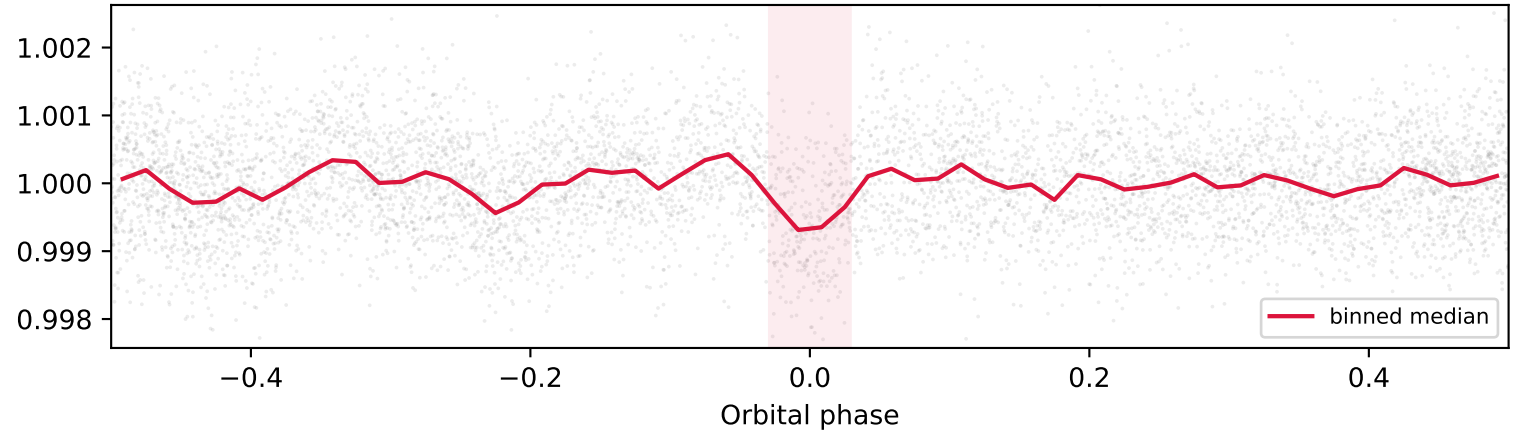
```
CATALOG CROSS-MATCH (novelty)

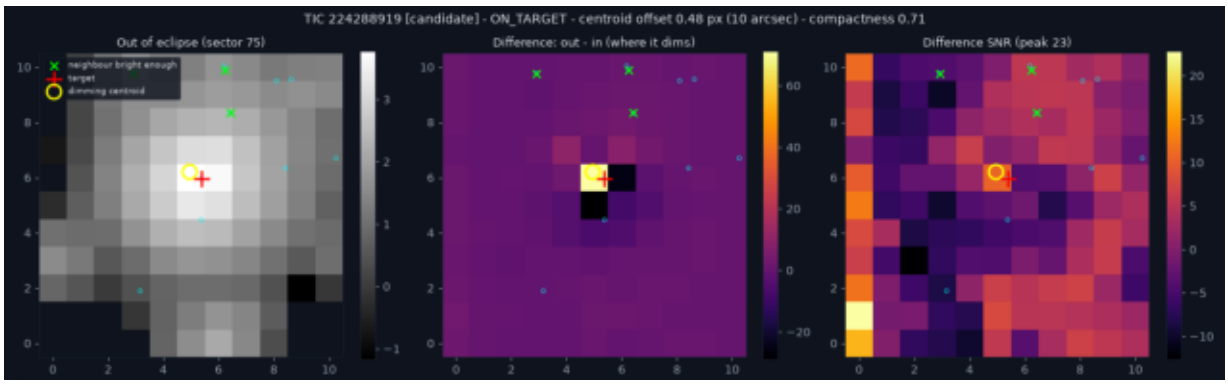
  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 75    light-curve dip in this sector: 10.6 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   22.89   (gate >= 5.0)      Compactness   0.71   (gate >= 0.50)
  Dimming centroid offset     0.48 px (10 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 224288920
    2.60 px away, dTmag +6.22 - farther from the dimming than the target
```

Phase-folded light curve (shaded = BLS eclipse window)





No caveats - clean detection on every check.